

Virtual parietal channels from a 2-channel frontal EEG: reconstructing P3/P4 and predicting arousal

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FACED dataset, 39 subjects · leave-subjects-out evaluation · September 2026

RQ1. Can a 2-channel frontal headset (F3/F4 or Fp1/Fp2) predict P3/P4 band power? **RQ2.** Can arousal be predicted from the frontal channels, the predicted (virtual) P3/P4, or both, compared with a real 4-channel montage? **RQ3.** Which electrodes carry arousal information, and how well do F3/F4 classify emotion categories?

Methods

Data. FACED (Chen et al., 2023; CC-BY-4.0), first 39 of 123 subjects: doi.org/10.82901/nemar.nm000112 (paper: doi.org/10.1038/s41597-023-02650-w). **Setup:** each participant watched 28 emotion-eliciting film clips (35–130 s each) while 32-channel EEG was recorded; after each clip they rated their own arousal and valence on 0–7 sliders. These self-ratings are our labels, one per clip (a trial); no video data is used, only EEG. We kept Fp1, Fp2, F3, F4, P3 and P4 (1–45 Hz band-pass, 50 Hz notch, resampled to 250 Hz) and cut each trial into 4 s windows with a 2 s step (34,314 windows).

Features. Per frontal channel: log power in five bands (δ 1–4, θ 4–8, α 8–13, β 13–30, γ 30–45 Hz), relative power, Hjorth mobility and complexity, spectral entropy and log variance (28 features per pair). An ablation also tested band power alone and band power + explicit ratios/asymmetries (β/α , θ/β , θ/α , right–left). All values z-scored within each subject (per-user calibration).

RQ1. Frontal features $\rightarrow$ P3/P4 log band power (10 targets), fitted with ridge regression and a neural network: a fully connected feed-forward net (scikit-learn MLP; 28 inputs $\rightarrow$ 256 $\rightarrow$ 128 ReLU units $\rightarrow$ 10 outputs; L2 penalty 0.01; Adam, learning rate 0.001, batch 256; early stopping on a 15% validation split). The same architecture with one output predicts arousal in RQ2. A no-learning baseline copies the same band from the same-side frontal electrode. Metrics: within-subject r between predicted and real power over time, and high/low accuracy (is P3/P4 power above or below that person's median in each window; chance 50%). **RQ2.** Trials were split at each subject's median arousal rating. Inputs (per trial): frontal features, virtual P3/P4 (out-of-subject RQ1 predictions), or both. References: real P3/P4, real 4-channel, self-reported valence, and the textbook β/α ratio with no fitting. Metrics: high/low accuracy (chance 50%) and within-subject r . **RQ3.** Per-trial log band power (5 bands) for all 30 scalp channels; every single channel and all 435 pairs (both channels + right–left difference) scored on high/low arousal with ridge. Luck check: 10 reruns with arousal shuffled within subject, keeping the best pair each time. Emotion categories (positive/negative/neutral; 9 clip categories) classified by logistic regression from F3/F4 or all 30 channels. All scores: 5-fold cross-validation by subject.

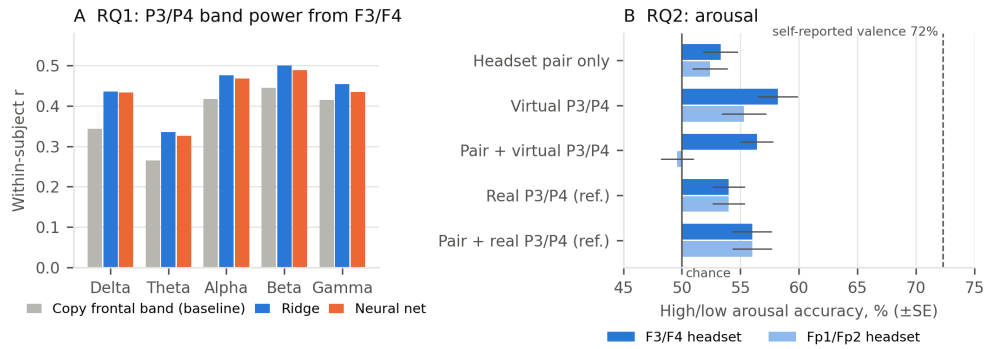


Figure 1. (A) RQ1 reconstruction accuracy per band, F3/F4 input, averaged over P3 and P4. (B) RQ2 high/low arousal accuracy from the neural net. Each “headset” is simulated by keeping only those two electrodes from the 32-channel recording; “headset pair”/“pair” means F3/F4 or Fp1/Fp2 according to bar colour; real P3/P4 is the same input for both.

Results

	Simulated F3/F4 headset	Simulated Fp1/Fp2 headset
RQ1 Band power r : copy baseline / ridge / neural net	0.38 / 0.44 / 0.43	0.26 / 0.32 / 0.31
RQ1 High/low accuracy: copy baseline / ridge / neural net	62% / 64% / 63%	58% / 60% / 60%
RQ1 r : band power only / + auxiliary (main) / + ratios (ridge)	0.43 / 0.44 / 0.44	0.31 / 0.32 / 0.32
RQ1 Alpha asymmetry P4–P3, r	0.04	0.02
RQ2 Arousal from headset pair only: accuracy (r)	53% (0.15)	52% (0.06)
RQ2 Arousal from virtual P3/P4	58% (0.18)	55% (0.18)
RQ2 Arousal from headset pair + virtual P3/P4	56% (0.17)	50% (0.08)
<i>Ref.</i> Arousal from real P3/P4 / headset pair + real P3/P4	54% (0.15) / 56% (0.20)	54% (0.15) / 56% (0.18)
<i>Ref.</i> β/α ratio, no model: r (accuracy)	–0.03 (49%)	–0.01 (50%)
<i>Ref.</i> Self-reported valence $\rightarrow$ arousal	72% (0.54)	72% (0.54)

Arousal accuracy (rank of 435 pairs)		
RQ3 Best pairs (F7–T5, FC5–T4, FC5–T5)	59.3% (1–3)	
RQ3 F3–F4 / P3–P4 / Fp1–Fp2	55.7% (145) / 54.8% (264) / 53.7% (356)	
RQ3 All 30 channels; shuffled-label best pair (max of 10)	57.9%; 55.7%	
	F3/F4 only	All 30 channels
RQ3 Positive / negative / neutral (chance 33%), balanced acc.	42%	56%
RQ3 9 emotion categories (chance 11%)	23%	42%

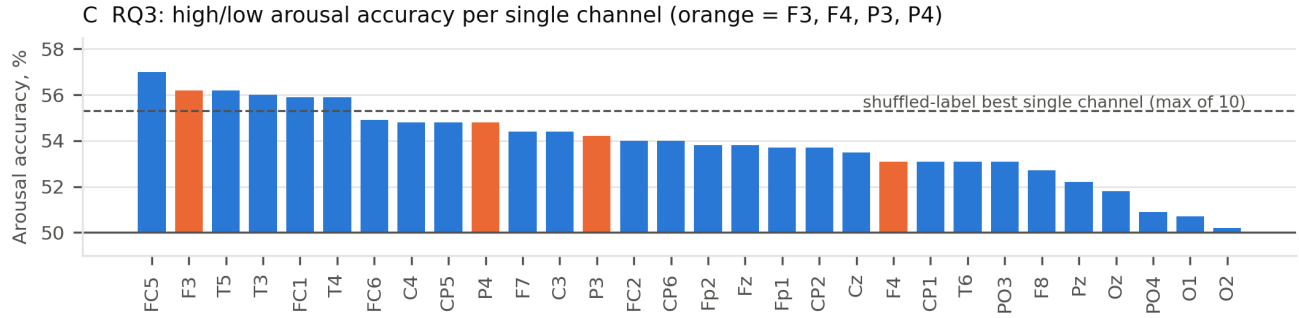


Figure 2. Single-channel high/low arousal accuracy (leave-subjects-out). Only channels above the dashed line beat the best result obtained with shuffled labels.

RQ1: yes, partly. F3/F4 predict P3/P4 band power at $r = 0.44$ in unseen subjects, i.e. 64% high/low accuracy (β and α best, θ worst), and a time-shuffle control gives $R^2 = 0.00$. Most of this is the part of parietal power shared with frontal sites: copying the frontal band already gives 0.38 (62%), and the neural net adds nothing over ridge. Auxiliary features add only +0.01 over band power alone, and explicit ratios add nothing (they are linear combinations of the log band powers already in the model). The parietal-only part, such as the P4–P3 asymmetry, is not recovered ($r \approx 0.03$). Fp1/Fp2 are weaker inputs ($r = 0.32$).

RQ2: weakly, and virtual P3/P4 help. With F3/F4, virtual P3/P4 gave the best EEG result (58% accuracy, $r = 0.18$), matching or exceeding frontal alone (53%), frontal + virtual (56%), real P3/P4 (54%) and real 4-channel (56%). Virtual channels contain no new information, so the gain likely comes from compressing the features; they match real P3/P4 probably because the arousal signal there is the shared part the reconstruction keeps. Frontal + virtual did not help and overfit with Fp1/Fp2 (50%). The β/α ratio carried no arousal information; frontal β power fell with arousal ($r = -0.17$), suggesting muscle or movement activity.

RQ3: P3/P4 is not a good arousal site; temporal + fronto-central is. The best pairs combine a left fronto-central electrode (F3, FC5, F7) with a temporal one (T3/T4/T5) and reach 59%, above the shuffled-label maximum (55.7%); 165 of 435 pairs reach at least that level. P3–P4 (54.8%) and F3–F4 (55.7%) do not. Across the scalp, alpha power falls with arousal ($r \approx -0.1$ to -0.16) and temporal gamma rises ($T4\ r = +0.17$; possibly partly muscle). All 30 channels reach only 58%. F3/F4 separate positive/negative/neutral (42% balanced vs 33% chance) and the 9 categories (23% vs 11%) above chance, but well below all 30 channels (56%, 42%); the 9-class score may partly reflect clip-specific responses, since all subjects saw the same clips.

Overall. F3/F4 can stand in for P3/P4 where overall parietal power matters (including arousal here), not for left–right measures. All EEG inputs stay far below self-reported valence (72%); the bottleneck is EEG arousal decoding, not the reconstruction. *Limitations:* 39 subjects (accuracy SE ≈ 1.5 –2 points); one dataset (film-clip induction); spectral features only.